

exercise5.py x

28

def nickname(self, new_nickname):

29

self._nickname = new_nickname

30

31

my_cyclist = Cyclist("Greg LeMond", "American", "Le Montstre")

32

print(my_cyclist.name)

33

print(my_cyclist.nationality)

34

print(my_cyclist.nickname)

35

my_cyclist.name = "Eddy Merckx"

36

my_cyclist.nationality = "Belgian"

37

my_cyclist.nickname = "The Cannibal"

38

print(my_cyclist.name)

39

print(my_cyclist.nationality)

40

print(my_cyclist.nickname)

Guide x

Collapse

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Encapsulation Exercise 5

Exercise 5

Define the `Cyclist` class which has attributes `name`, `nationality`, and `nickname`. Use the Python convention to make these attributes private. Create the getters and setters using the `property` decorator.

Expected Output

Initialize an object of the `Cyclist` class as follows:

```
my_cyclist = Cyclist("Greg LeMond", "American", "Le Montstre")
```

Task	Output
<code>print(my_cyclist.name)</code>	<code>Greg LeMond</code>
<code>print(my_cyclist.nationality)</code>	<code>American</code>
<code>print(my_cyclist.nickname)</code>	<code>Le Montstre</code>
<code>my_cyclist.name = "Eddy Merckx"</code>	N/A
<code>my_cyclist.nationality = "Belgian"</code>	N/A
<code>my_cyclist.nickname = "The Cannibal"</code>	N/A
<code>print(my_cyclist.name)</code>	<code>Eddy Merckx</code>
<code>print(my_cyclist.nationality)</code>	<code>Belgian</code>
<code>print(my_cyclist.nickname)</code>	<code>The Cannibal</code>

Important

Use the property decorator; follow the Python convention for private attributes.

TRY IT

Greg LeMond
American
Le Montstre
Eddy Merckx
Belgian
The Cannibal

Submit your code when ready.

Here is one possible solution:

```
class Cyclist:
    def __init__(self, name, nationality, nickname):
        self._name = name
        self._nationality = nationality
        self._nickname = nickname

    @property
    def name(self):
        return self._name

    @name.setter
    def name(self, new_value):
        self._name = new_value

    @property
    def nationality(self):
        return self._nationality

    @nationality.setter
    def nationality(self, new_value):
        self._nationality = new_value

    @property
    def nickname(self):
        return self._nickname

    @nickname.setter
    def nickname(self, new_value):
        self._nickname = new_value
```

Check It!

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Test passed

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